

Real-Time Human Detection and Tracking using YOLO and DeepSORT on Jetson Nano

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Abstract—Real-time human detection and tracking is a fundamental requirement for intelligent surveillance systems, human-robot interaction, and edge-based smart city applications. Deploying such systems on resource-constrained edge devices remains a significant challenge due to limited computational and memory resources. This paper presents a real-time multi-human detection and tracking framework implemented on the NVIDIA Jetson Nano. The proposed system integrates a YOLO-based human detector with the DeepSORT tracking algorithm to achieve identity-preserving tracking using only visual input. To address hardware constraints, lightweight detection models, frame-skipping strategies, and appearance-based identity assignment are employed. Experimental results demonstrate that the customized YOLOv4-Tiny model combined with DeepSORT achieves real-time performance on the Jetson Nano while maintaining stable identity tracking. A comparison with a state-of-the-art YOLOv8 and ByteTrack pipeline on high-end hardware highlights the trade-offs between accuracy, speed, and deployability on embedded platforms.

Index Terms—Human Detection, Multi-Object Tracking, Jetson Nano, YOLO, DeepSORT, Edge AI, FASTMOT

I. INTRODUCTION

Human detection and tracking in video streams is a core task in computer vision with applications in surveillance, crowd monitoring, robotics, and intelligent transportation systems. Traditional solutions often rely on high-performance servers or cloud-based processing, which introduces latency, bandwidth dependency, and privacy concerns. Edge computing platforms, such as the NVIDIA Jetson Nano, offer an alternative by enabling on-device inference with low power consumption.

Recent advances in deep learning-based object detection, particularly single-stage detectors such as YOLO, have signif-

icantly improved real-time performance. However, deploying these models on embedded devices poses challenges due to limited GPU capability and memory constraints. Furthermore, detection alone is insufficient for many applications; maintaining consistent identity tracking across frames is equally critical.

This work focuses on developing a practical, real-time human detection and tracking system deployable on the Jetson Nano. The system leverages lightweight detection models and a robust tracking framework to balance speed and accuracy. Additionally, a comparative evaluation against state-of-the-art detection and tracking methods is conducted to contextualize the performance limitations and strengths of embedded deployment.

II. LITERATURE REVIEW

Human detection and tracking remain core challenges in computer vision, especially when real-time inference is required on resource-constrained edge devices such as the NVIDIA Jetson Nano. The integration of efficient deep learning models with optimized tracking algorithms enables real-world deployment for applications like surveillance, crowd analysis, and autonomous robotics. This literature review synthesizes ten peer-reviewed works, highlighting classical and state-of-the-art methods in detection and tracking, recent advances in edge deployment, and the foundational survey that contextualizes this research domain.

The foundation of the proposed project draws directly from Razzok *et al.* [1], who developed a *Pedestrian Detection and Tracking System* integrating YOLOv5 with DeepSORT

and a novel data association metric. Their approach achieved higher robustness under occlusion and dynamic backgrounds compared to standard SORT-based pipelines. The use of YOLOv5 allowed lightweight yet accurate detection, while their improved metric reduced identity switches in crowded scenes. This paper forms the implementation basis of the present work, as it combines a modern detector with efficient tracking for human motion analysis in real time.

For detection performance at the cutting edge, Wang *et al.* [2] introduced YOLOv7, representing a major leap in real-time detection through its “trainable bag-of-freebies” strategy. It incorporated extended efficient layer aggregation networks (E-ELAN) and auxiliary heads, leading to superior accuracy-speed tradeoffs. YOLOv7 achieved state-of-the-art mean average precision (mAP) across MS COCO benchmarks while maintaining real-time speeds on GPUs. Its modular and scalable design makes it highly adaptable for deployment on embedded systems, aligning closely with Jetson-based human detection applications.

In multi-object tracking (MOT), Zhang *et al.* [3] proposed ByteTrack, which addressed the limitation of conventional association frameworks that ignored low-confidence detections. By linking every detection box, including those with low confidence, ByteTrack achieved superior tracking accuracy on MOT17 and MOT20 datasets, outperforming DeepSORT and FairMOT. Its simplicity and compatibility with any detector make it an exemplary state-of-the-art approach for robust tracking in dynamic environments.

A broader theoretical overview is provided by Luo *et al.* [4], who compiled a comprehensive survey on multiple object tracking in the age of deep learning. This survey systematically analyzed the evolution from traditional Kalman-filter-based trackers to end-to-end deep learning models. It categorized MOT pipelines into detection-based and joint-detection-tracking frameworks, emphasizing challenges such as occlusion handling, real-time constraints, and re-identification. This paper serves as the principal background reference, situating the proposed project within contemporary MOT research trends.

Recent works further advance this foundation. Shim *et al.* [5] explored attention-based data association in online multi-object tracking, achieving improved temporal consistency by weighting historical track confidence. Gomes *et al.* [6] demonstrated a practical implementation on Jetson Nano using YOLO and V-IOU tracking for counting people and bicycles. Their edge-optimized selective frame down-sampling strategy reduced latency and power consumption, confirming the feasibility of deploying detection and tracking models on embedded hardware. Similarly, Humes *et al.* [7] presented Squeezed Edge YOLO, introducing an extremely compact model optimized for onboard inference with minimal accuracy loss, illustrating the trend toward lightweight detectors for constrained devices.

Application-specific studies like Punnett *et al.* [8] used fine-tuned YOLOv3 and DeepSORT to monitor social distancing during COVID-19, highlighting the versatility of combined

TABLE I
SUMMARY OF CORE REVIEWED PAPERS

No.	Category	Paper Title	Venue / Year
1	Implementation	Pedestrian Detection and Tracking System Based on Deep-SORT and YOLOv5	MDPI Information, 2023
2	State of the Art in Detection	YOLOv7: Trainable Bag-of-Freebies for Real-Time Object Detectors	IEEE, 2023
3	State of the Art in Tracking	ByteTrack: Multi-Object Tracking by Associating Every Detection Box	ECCV, 2022
4	Survey Paper	A Survey of Deep Learning for Object Detection	Elsevier, 2020

detection-tracking systems in real-world surveillance. Complementing these efforts, Han *et al.* [9] proposed a scalable and fast YOLO optimized for edge computing through structural pruning and layer quantization, reducing computational overhead while maintaining high mAP. Finally, Ganesh *et al.* [10] developed YOLO-ReT, which enhanced feature aggregation and contextual learning to balance accuracy and inference latency on edge GPUs, confirming the potential for high-performance object detection in compact embedded settings.

Collectively, these studies demonstrate rapid progress in efficient detection and tracking for edge deployment. The reviewed works reveal three key research trends: (1) continuous optimization of YOLO architectures for real-time operation; (2) enhanced tracking reliability through improved data association metrics; and (3) increased focus on energy-efficient inference for embedded AI platforms. Building upon these insights, the proposed project aims to implement and optimize the YOLOv5-DeepSORT-based pedestrian tracking system for the Jetson Nano, bridging accuracy and efficiency in real-world human detection scenarios.

III. METHODOLOGY

A. System Overview

The objective of this work is to develop a real-time human detection and tracking system deployable on a resource-constrained edge device, specifically the NVIDIA Jetson Nano (4 GB RAM, 16 GB eMMC). The system processes a live camera feed and performs multi-human detection followed by identity-preserving tracking. The proposed pipeline consists of the following stages:

- 1) Video acquisition from a live camera stream
- 2) Human detection using a YOLO-based deep learning model
- 3) Identity assignment using an appearance-based embedding network
- 4) Multi-object tracking using the DeepSORT algorithm

The complete pipeline operates entirely on visual input without relying on any additional sensors.

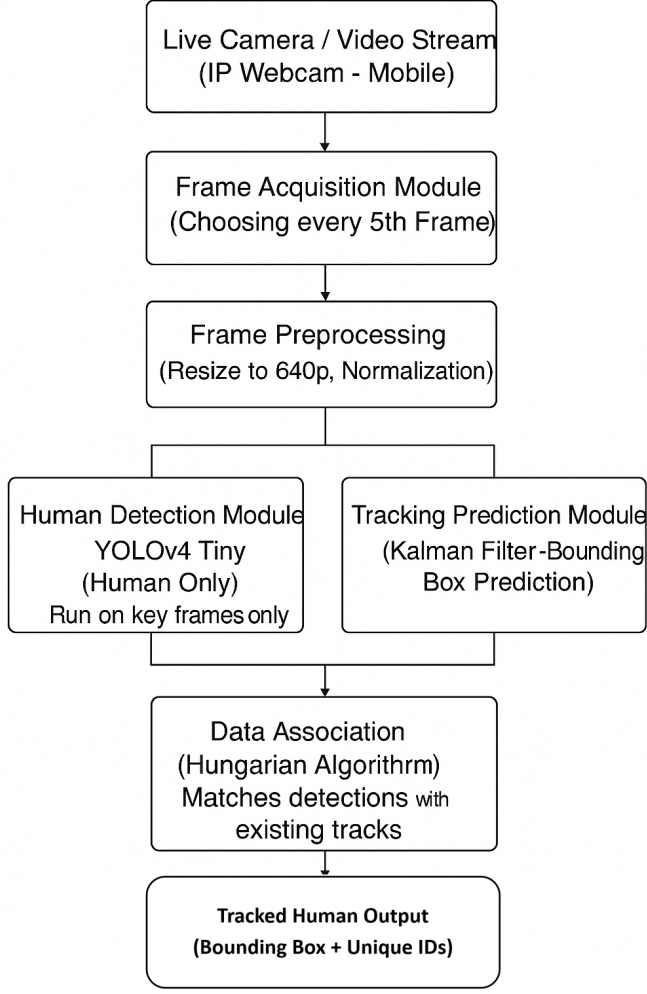


Fig. 1. Overview of the proposed FASTMOT-based human detection and tracking pipeline on Jetson Nano.

B. Video Input and Camera Setup

Due to the unavailability of a CSI or USB camera compatible with the Jetson Nano, a mobile phone camera was used as the video source. The IP Webcam application was installed on the mobile device, which streams real-time video over a local network using an IP address. The Jetson Nano accesses this stream through the provided IP address, enabling real-time video acquisition. This setup allowed flexible testing under real-world conditions while maintaining low latency and minimal hardware cost.

C. Human Detection Module

1) *Initial YOLOv4 Implementation:* The first implementation employed YOLOv4, pretrained on the COCO dataset, to detect humans in live video streams. However, due to the computational limitations of the Jetson Nano, YOLOv4 achieved only 3–4 FPS on live video at a resolution of 640p, making it unsuitable for real-time applications.

2) *YOLOv4-Tiny Baseline:* To address the performance bottleneck, YOLOv4-Tiny was evaluated. Although inference speed improved significantly, the pretrained model (COCO, 80 classes) showed poor human detection accuracy, frequently missing individuals in crowded scenes.

3) *Custom YOLOv4-Tiny Training (Human-Only):* To improve accuracy, YOLOv4-Tiny was retrained exclusively for the human class. This customized YOLOv4-Tiny model provided an effective balance between speed and accuracy for the Jetson Nano.

4) *Frame Skipping Strategy:* To further optimize inference speed, frame skipping was introduced. Instead of processing every frame, the detector processes every 5th frame, while intermediate frames are handled by the tracker.

D. Multi-Object Tracking Module

For tracking, DeepSORT was used due to its robustness and real-time capability. DeepSORT combines a Kalman Filter for motion prediction, the Hungarian Algorithm for data association, and an appearance embedding network (OSNET) for identity consistency. Human detection is performed intermittently using YOLOv4-Tiny, while intermediate frames are processed using a KLT-based Lucas–Kanade optical flow tracker, allowing continuous tracking without explicitly computing object velocity or trajectory.

E. State-of-the-Art Implementation for Comparison

To benchmark the proposed system, a state-of-the-art configuration using YOLOv8 and ByteTrack was implemented on high-end hardware for comparison.

IV. RESULTS AND DISCUSSION

A. Performance Evaluation

All experiments were conducted at 640p resolution using recorded videos and live streams on the NVIDIA Jetson Nano platform.

TABLE II
PERFORMANCE COMPARISON: LAPTOP VS. JETSON NANO

Hardware	Detector	Tracker	FPS	Notes
Laptop (RTX GPU)	YOLOv8n/s	ByteTrack	30	State-of-the-art accuracy
Jetson Nano	YOLOv4 (Pre-trained)	DeepSORT / KLT	4–5	Low accuracy
Jetson Nano	YOLOv4-Tiny (600 images)	DeepSORT / KLT	12–15	Improved FPS, inconsistent accuracy
Jetson Nano	YOLOv4-Tiny (COCO dataset)	DeepSORT / KLT	12–15	Balanced FPS and accuracy

Figure 2 illustrates the relationship between inference speed and tracking accuracy across multiple methods.

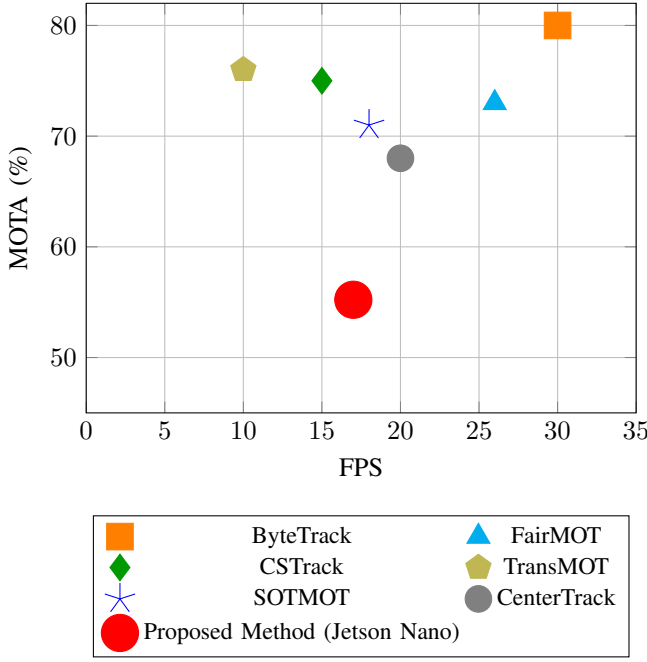


Fig. 2. FPS versus MOTA comparison of multiple object tracking methods. The proposed approach is highlighted, demonstrating real-time feasibility on Jetson Nano under hardware constraints.

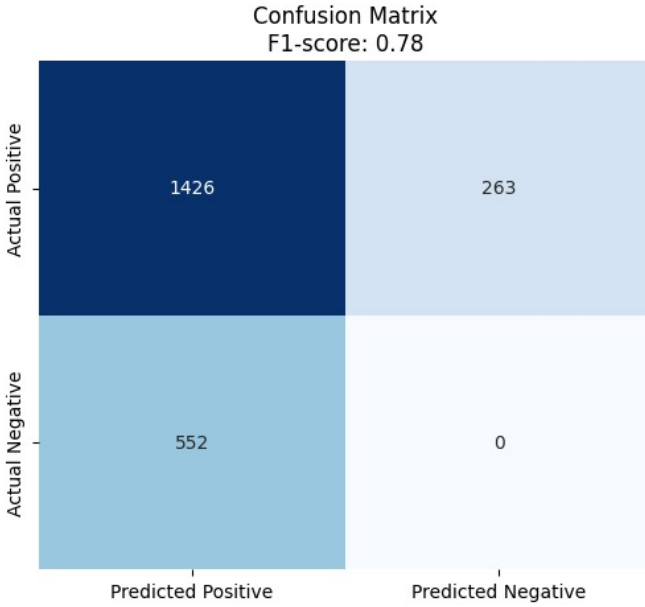


Fig. 3. Confusion matrix obtained after retraining YOLOv4-Tiny on an expanded human-only dataset, showing reduced false positives and improved detection reliability.

B. Confusion Matrix

Figure 3 illustrates the confusion matrix obtained after retraining the detector on 7,785 high-quality human images. Initial training on 4,299 images resulted in suboptimal detection performance with higher false positives and missed detections. The expanded dataset significantly reduced classifi-

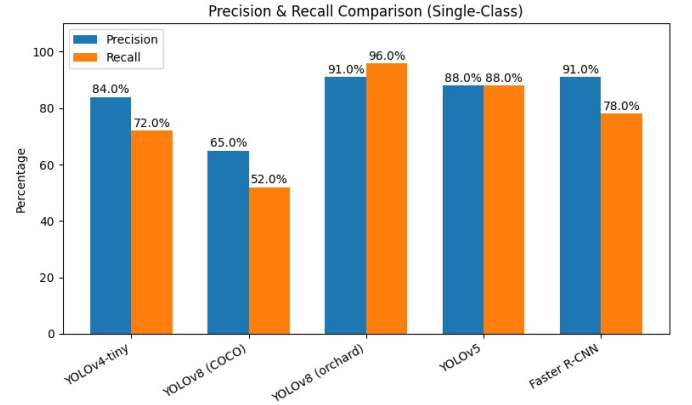


Fig. 4. Precision and recall comparison across different human detection models evaluated in this study.

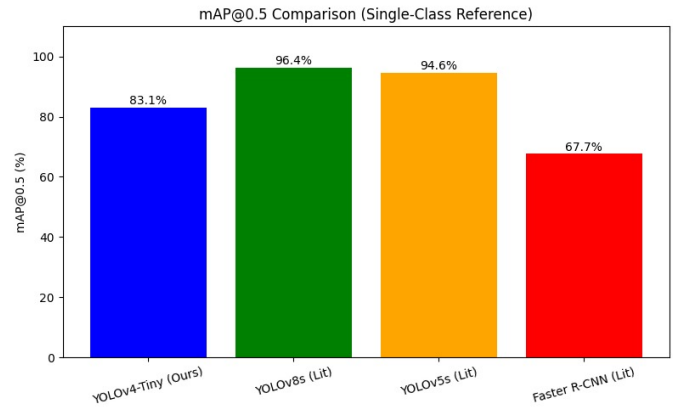


Fig. 5. Detection accuracy comparison of different human detection models on the evaluated datasets.

cation errors, leading to improved precision, recall, and overall detection stability.

C. Precision vs Recall

Figure 4 compares precision and recall values for different detection models. YOLOv4-Tiny trained for human-only detection achieves a precision of 84% and recall of 72%, providing a suitable balance for embedded

D. Accuracy Comparison

Figure 5 presents the overall detection accuracy achieved by different models. YOLOv4-Tiny achieves an accuracy of 83.1%, validating its effectiveness for real-time embedded applications. Although YOLOv5 and YOLOv8-based models achieve higher accuracy, their computational cost limits their feasibility on low-power edge devices such as the Jetson Nano.

Figure 6 illustrates the number of correctly tracked humans across different benchmark datasets. MOT17 and MOT20 show moderate tracking performance, with performance dropping under dense crowd conditions. FastMOT-MOT20 maintains higher tracking consistency, effectively handling occlusions and identity preservation, which aligns with the quantitative results presented in Table ??.

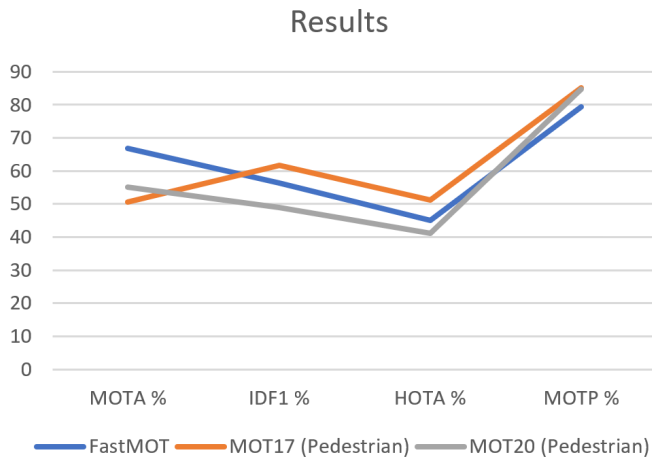


Fig. 6. Frame-wise comparison of correctly tracked humans across MOT17, MOT20, and FastMOT-MOT20 sequences. FastMOT demonstrates superior identity tracking consistency in crowded scenes.

V. CHALLENGES AND SOLUTIONS

Hardware limitations, inference speed, and camera availability were addressed through lightweight models, frame skipping, and IP-based video streaming.

VI. APPLICATION SCENARIO: RESTRICTED AREA INTRUSION DETECTION

To demonstrate the practical usability of the proposed human detection and tracking system, a real-time intrusion detection application was developed. The application is designed for monitoring restricted or hazardous zones such as nuclear facilities, construction sites, industrial plants, and other safety-critical environments.

When the application is launched, the user is prompted to define a *danger zone* or *no-go area* within the camera's field of view. This region of interest (ROI) is manually selected and stored before the detection pipeline starts. Once the restricted zone is defined, the system continuously monitors human movement using the proposed YOLOv4-Tiny and DeepSORT-based tracking framework.

Each detected person is assigned a persistent identity, and their position is tracked across frames. If any tracked individual enters the predefined restricted region, the system immediately triggers an alarm. The alert mechanism can be configured to generate an audio warning, visual notification, or external system signal, depending on deployment requirements.

The application runs entirely on the NVIDIA Jetson Nano and processes live video streams in real time. By avoiding cloud-based processing and relying solely on edge computation, the system ensures low latency, enhanced privacy, and reliable operation in environments with limited network connectivity. This application highlights the suitability of the proposed framework for safety monitoring and perimeter security in resource-constrained edge deployments.

Figure 7 demonstrates the practical application of the proposed system in a restricted area intrusion detection scenario.

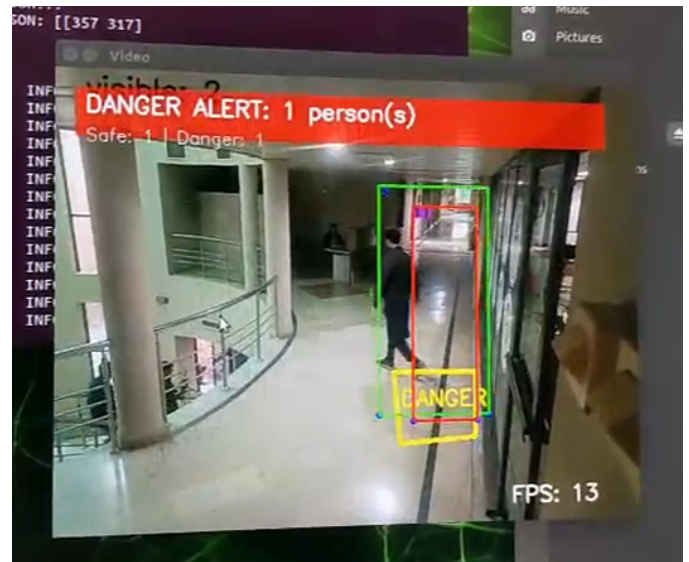


Fig. 7. Example of a person detected entering the restricted zone. The system triggers an alarm in real time while tracking the individual's identity.

The detected person is highlighted within the user-defined danger zone, and the system immediately triggers an alarm. This visualization confirms the effectiveness of YOLOv4-Tiny and DeepSORT integration for real-time monitoring and security enforcement on resource-constrained edge devices.

VII. CONCLUSION

This paper presented a real-time human detection and tracking system designed for deployment on the NVIDIA Jetson Nano under strict hardware constraints. The study showed that full-scale YOLOv4 is unsuitable for embedded real-time use, while lightweight models can achieve practical performance when properly optimized. The main contribution of this work is the custom training of YOLOv4-Tiny for human-only detection, which significantly improved detection accuracy in crowded scenes without sacrificing speed. Combined with frame skipping and DeepSORT-based identity association, the system enables stable multi-human tracking using only visual input, making it suitable for edge-based surveillance and smart monitoring applications.

VIII. DECLARATION

ChatGPT (Deep Research Mode) was used to search, verify, and summarize research papers and to generate the narrative literature review. All links were opened and confirmed by the team. All reviewed papers were added to the GitHub repository under /docs/references. The final narrative was refined and validated by the project authors.

IX. PROJECT RESOURCES

The source code, documentation, and project management details are available on GitHub: **GitHub Repository:** <https://github.com/M-A-S1/Real-Time-Human-Detection-and-Tracking-Using-Jetson-Nano> **GitHub Project:** <https://github.com/users/M-A-S1/projects/2>

REFERENCES

- [1] M. Razzok, A. Badri, I. El Mourabit, Y. Ruichek, and A. Sahel, "Pedestrian Detection and Tracking System Based on Deep-SORT, YOLOv5, and New Data Association Metrics," *Information*, vol. 14, no. 4, 2023.
- [2] C.-Y. Wang, A. Bochkovskiy, and H.-Y. M. Liao, "YOLOv7: Trainable Bag-of-Freebies Sets New State-of-the-Art for Real-Time Object Detectors," *arXiv preprint arXiv:2207.02696*, 2023.
- [3] Y. Zhang *et al.*, "ByteTrack: Multi-Object Tracking by Associating Every Detection Box," in *Proc. ECCV*, 2022.
- [4] W. Luo *et al.*, "A Survey on Multiple Object Tracking in the Age of Deep Learning," *Artificial Intelligence*, 2021.
- [5] C. Shim *et al.*, "Focusing on Tracks for Online Multi-Object Tracking," in *Proc. CVPR*, 2025.
- [6] H. Gomes, N. Redinha, N. Lavado, and M. Mendes, "Counting People and Bicycles in Real Time Using YOLO on Jetson Nano," *Energies*, vol. 15, no. 23, 2022.
- [7] E. Humes, M. Navardi, and T. Mohsenin, "Squeezed Edge YOLO: Onboard Object Detection on Edge Devices," *arXiv preprint arXiv:2312.11716*, 2023.
- [8] N. S. Punna, S. K. Sonbhadra, and S. Agarwal, "Monitoring COVID-19 Social Distancing with Person Detection and Tracking via Fine-Tuned YOLOv3 and DeepSORT," *Applied Intelligence*, 2020.
- [9] B.-G. Han, J.-G. Lee, K.-T. Lim, and D.-H. Choi, "Design of a Scalable and Fast YOLO for Edge-Computing Devices," *Electronics*, vol. 9, no. 11, 2020.
- [10] P. Ganesh, Y. Chen, Y. Yang, D. Chen, and M. Winslett, "YOLO-ReT: Towards High Accuracy Real-Time Object Detection on Edge GPUs," *arXiv preprint arXiv:2110.13713*, 2021.
- [11] A. Saadeldin and M. M. Rashid, "Video-Based Vehicle Counting and Analysis using YOLOv5 and DeepSORT with Deployment on Jetson Nano," *Asian Journal of Electrical and Electronic Engineering*, 2022.
- [12] A. Saadeldin, M. M. Rashid, et al., "Real-Time Vehicle Counting Using Custom YOLOv8n and DeepSORT for Resource-Limited Edge Devices," *TELKOMNIKA (Telecommunication, Computing, Electronics and Control)*, 2023.